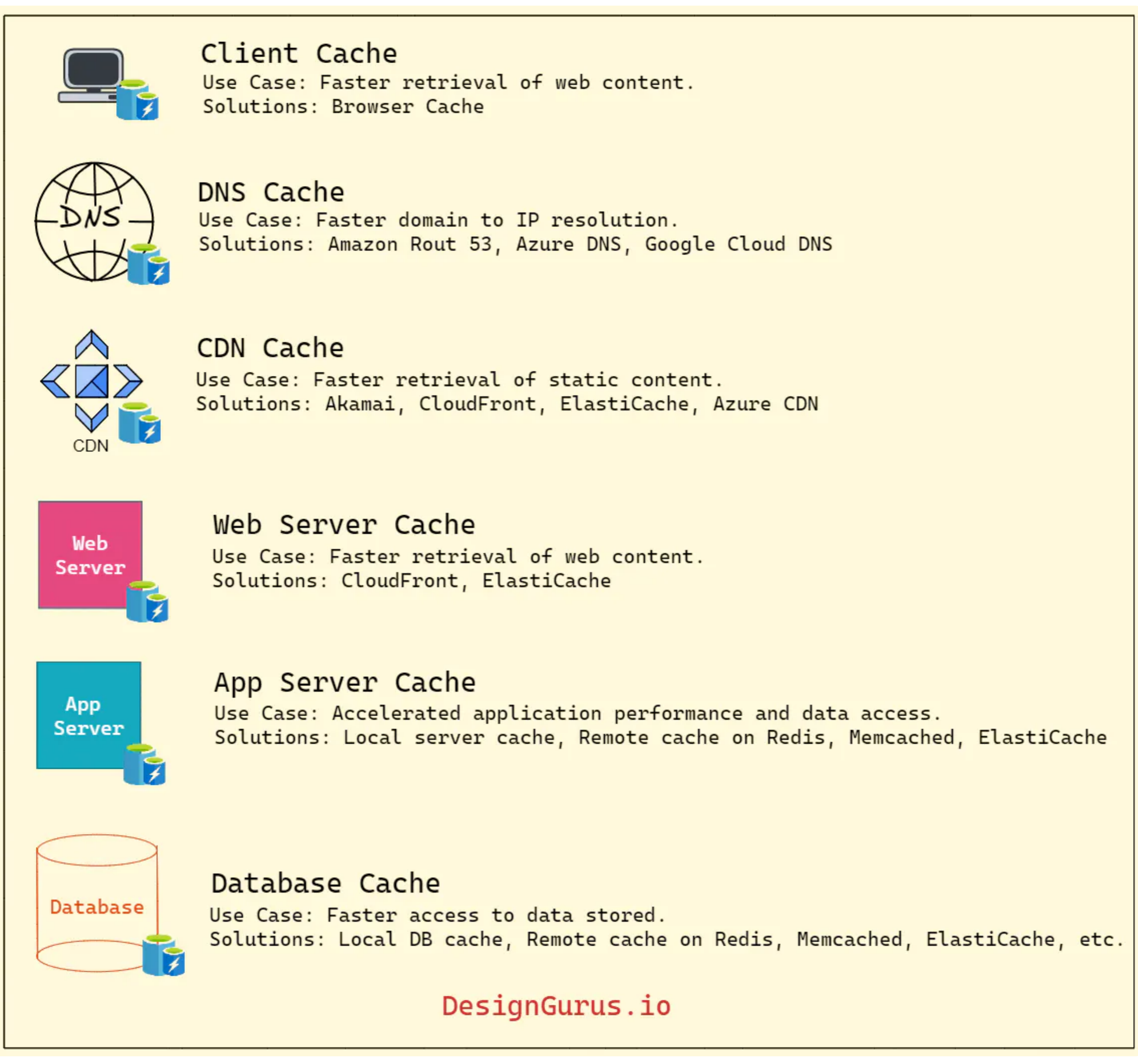
Types of Caching

Caching can be implemented in various ways, depending on the specific use case and the type of data being cached. Here are some of the most common types of caching:



Client Cache
Use Case: Faster retrieval of web content.
Solutions: Browser Cache

DNS Cache
Use Case: Faster domain to IP resolution.
Solutions: Amazon Rout 53, Azure DNS, Google Cloud DNS

CDN Cache
Use Case: Faster retrieval of static content.
Solutions: Akamai, CloudFront, ElastiCache, Azure CDN

Web Server Cache
Use Case: Faster retrieval of web content.
Solutions: CloudFront, ElastiCache

App Server Cache
Use Case: Accelerated application performance and data access.
Solutions: Local server cache, Remote cache on Redis, Memcached, ElastiCache

Database Cache
Use Case: Faster access to data stored.
Solutions: Local DB cache, Remote cache on Redis, Memcached, ElastiCache, etc.

DNS
CDN
Web Server
App Server
Database
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1. In-memory caching

In-memory caching stores data in the main memory of the computer, which is faster to access than disk storage. In-memory caching is useful for frequently accessed data that can fit into the available memory. This type of caching is commonly used for caching API responses, session data, and web page fragments. To implement in-memory caching, software engineers can use various techniques, including using a cache library like Memcached or Redis, or implementing custom caching logic within the application code.

2. Disk caching

Disk caching stores data on the hard disk, which is slower than main memory but faster than retrieving data from a remote source. Disk caching is useful for data that is too large to fit in memory or for data that needs to persist between application restarts. This type of caching is commonly used for caching database queries and file system data.

3. Database caching

Database caching stores frequently accessed data in the database itself, reducing the need to access external storage. This type of caching is useful for data that is stored in a database and frequently accessed by multiple users. Database caching can be implemented using a variety of techniques, including database query caching and result set caching.

4. Client-side caching

This type of caching occurs on the client device, such as a web browser or mobile app. Client-side caching stores frequently accessed data, such as images, CSS, or JavaScript files, to reduce the need for repeated requests to the server. Examples of client-side caching include browser caching and local storage.

5. Server-side caching

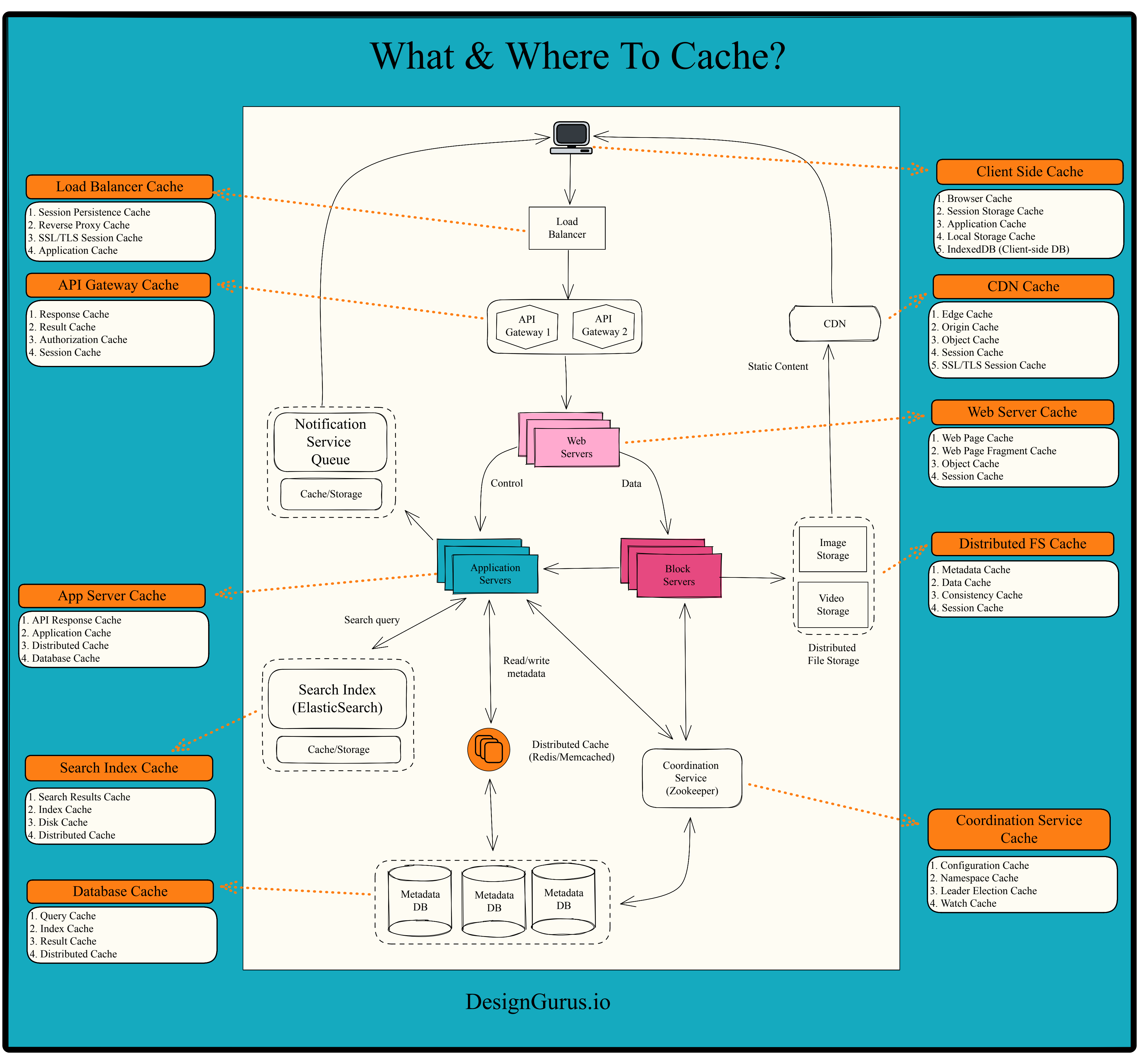
This type of caching occurs on the server, typically in web applications or other backend systems. Server-side caching can be used to store frequently accessed data, precomputed results, or intermediate processing results to improve the performance of the server. Examples of server-side caching include full-page caching, fragment caching, and object caching.

6. CDN caching

CDN caching stores data on a distributed network of servers, reducing the latency of accessing data from remote locations. This type of caching is useful for data that is accessed from multiple locations around the world, such as images, videos, and other static assets. CDN caching is commonly used for content delivery networks and large-scale web applications.

7. DNS caching

DNS cache is a type of cache used in the Domain Name System (DNS) to store the results of DNS queries for a period of time. When a user requests to access a website, their computer sends a DNS query to a DNS server to resolve the website's domain name to an IP address. The DNS server responds with the IP address, and the user's computer can then access the website using the IP address. DNS caching improves the performance of the DNS system by reducing the number of requests made to DNS servers. When a DNS server receives a request for a domain name, it checks its local cache to see if it has the IP address for that domain name. If the IP address is in the cache, the DNS server can immediately respond with the IP address without having to query other servers. This can significantly reduce the response time for DNS queries and improve the overall performance of the system.



What & Where To Cache?
Load Balancer Cache
1. Session Persistence Cache
2. Reverse Proxy Cache
3. SSL/TLS Session Cache
4. Application Cache
API Gateway Cache
1. Response Cache
2. Result Cache
3. Authorization Cache
4. Session Cache
App Server Cache
1. API Response Cache
2. Application Cache
3. Distributed Cache
4. Database Cache
Search Index Cache
1. Search Results Cache
2. Index Cache
3. Disk Cache
4. Distributed Cache
Database Cache
1. Query Cache
2. Index Cache
3. Result Cache
4. Distributed Cache
Load Balancer
API Gateway 1
API Gateway 2
CDN
Static Content
Notification Service Queue
Cache/Storage
Web Servers
Control
Data
Application Servers
Block Servers
Image Storage
Video Storage
Distributed File Storage
Search query
Read/write metadata
Search Index (ElasticSearch)
Cache/Storage
Distributed Cache (Redis/Memcached)
Coordination Service (Zookeeper)
Metadata DB
Metadata DB
Metadata DB
Client Side Cache
1. Browser Cache
2. Session Storage Cache
3. Application Cache
4. Local Storage Cache
5. IndexedDB (Client-side DB)
CDN Cache
1. Edge Cache
2. Origin Cache
3. Object Cache
4. Session Cache
5. SSL/TLS Session Cache
Web Server Cache
1. Web Page Cache
2. Web Page Fragment Cache
3. Object Cache
4. Session Cache
Distributed FS Cache
1. Metadata Cache
2. Data Cache
3. Consistency Cache
4. Session Cache
Coordination Service Cache
1. Configuration Cache
2. Namespace Cache
3. Leader Election Cache
4. Watch Cache
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Why Is Caching Important?

Next →
Cache Replacement Policies

☐ Mark as Completed